ZOO501 (MCQS)

PREPARE BY: MAAH WARRIOR

TOPIC NO.1

1. -----which divide mitotically to produce all the cells of the body____ **Zygote**

TOPIC NO.3

2. -----the study of birth defect____ **Teratology**

TOPIC NO.4

3. A condition in which cell lack the instructions for development____ **naïve**

TOPIC NO.5

4. Strategies of commitment_____**2**

TOPIC NO.6

5. Mode of commitment_____ **3**

TOPIC NO.8

6. Conditional specification gives rise to a pattern of embryogenesis____ **regulative development.**
7. Regulative development is seen in most vertebrate embryos, and it is obviously------ of identical twins_____ **critical in the development**
8. The ability of the embryonic cell to change their fates to compensate for the missing part called _____ **Regulation**

TOPIC NO.9

9. Insects such as----of specification to commit their cells to particular fates____ **Drosophila use all three modes**

TOPIC NO.12

10. In amphibians, metamorphosis is initiated by hormones from the tadpole's____ **thyroid gland**
11. The speed of metamorphosis is directly linked with____ **environment pressures**

TOPIC NO.13

12. Rana pipiens usually lay around-----eggs, while the bullfrog, Rana catesbiana, can lay ----____**2500 & 20,000.**

TOPIC NO.15

13.------exchanged nuclei b/w two morphologically distinct species____ **J.Hammerling**

14.Development us controlled at both the ____ **transcriptional and translational level**

TOPIC NO.17

15.The individuals of each species, however is divided into two mating types Plus and minus____ **Chlamydomonas**

TOPIC NO.25

16.Nucleosome is composed of an octamer of histone proteins wrapped with two loops containing approximately------of DNA_____**140 base pairs**

TOPIC NO.26

17.Allows the mRNA to exit the nucleus_____ **poly(A) tail**

TOPIC NO.27-28

18.Promoter is enhancer which is thought to _____**regulate transcription**

TOPIC NO.34-35

19.The first evidence suggesting the importance of sperm in reproduction came from a series of experiments performed by_____ **Lazzaro Spallanzani**

TOPIC NO.38

20.It is estimated that the eggs of sea urchins contain----- different types of mRNA____**25,000 to 50,000**

21.The volume of a sea urchin egg is about____ **200 picoliters**

TOPIC NO.40

22.During fertilization, these actin molecules polymerize to form long cables of actin known as_____ **microfilaments**

23.Vitelline envelop contains at least eight different ___ **glycoproteins.**

TOPIC NO.48

24.Using immunological techniques,….demonstrated that bindin is located specifically on the acrosomal process exactly where it should be for sperm-egg recagnition?____**Moy and Vacquier**

25.The acrosomal protein mediating this recognition is called bindin. In……..vacquier and co-workers isolated this nonsoluble 30.500-Ds (Dalton) protein from the acrosome of Strongylacentrotus purpuratus and found it to be capable of binding to dejellied eggs of the same species?_____**1977**

TOPIC NO.47

26. Bleil and Wassarman (_____) isolated an 83-kDa glycoprotein, ZP3, from the mouse zona that was the active site for sperm binding. The other two zona glycoproteins they found, ZP1 and ZP2, failed to compete for sperm binding?_____ **1988**

27. The ____ is the specific glycoprotein in the mouse zona pellucida to which the sperm bind. _____ **ZP3**

28. The _____ is a glycoprotein matrix, which is synthesized and secreted by the growing oocyte?_____ **Zona pellucida**

TOPIC NO.48

29. The difference between the acrosomal reaction in sea urchins and mammals may be due to the thickness of the extracellular envelopes surrounding the egg. In the sea urchin, the vitelline envelope is very thin and porous?_____ **True**

30. It appears that the acrosome-reacted sperm transfer their binding from ZP3 to the adjacent _____ molecules. After a mouse sperm has entered the egg, the egg cortical granules release their contents._____ **ZP2**

TOPIC NO.49-50

31. The set of physiological changes that allow the sperm to be competent to fertilize the egg is called capacitation. The requirement for capacitation varies from species to species?_____ **True**

32. The Wilcox and colleagues…. found that nearly all human pregnancies result from sexual intercourse during a 6-day period ending on the day of ovulation?_____**1995**

33. The molecular changes that account for capacitation are of _____types?_____**4**

34. In capacitation, the ……. the fluidity of the sperm plasma membrane is altered by the removal of cholesterol by albumin proteins found in the female reproductive tract? _____**First**

35. In capacitation, the _____, particular proteins or carbohydrates on the sperm surface are lost during capacitation?_____ **Second**

36. In capacitation, the _____, the membrane potential of the sperm becomes more negative as potassium ions leave the sperm?_____ **Third**

37. In capacitation, the _____, protein phosphorylation occurs?_____ **Fourth**

TOPIC NO.51-52

38. Recognition of sperm by the vitelline envelope or zona pellucida is followed by the lysis of that portion of the envelope or zona in the region of the sperm head by the enzymes? ____ **Acrosomal**

39. The sperm and egg plasma membranes then join together, and material from the sperm membrane can later be found on the egg membrane. The sperm nucleus and tail pass

through the resulting cytoplasmic bridge, which is widened by the ____?**Actin polymerization**

40. In mammals, the _____ proteins in the sperm plasma membrane are essential for sperm membrane-egg membrane fusion?_____ **Fertilin**

41. The Ferilin _____ the sperm to the egg by binding to the 6β1 integrin protein on the egg plasma membrane. Moreover, like sea urchin bindin (to which it is not structurally related), fertilin has a hydrophobic region that could potentially mediate the union of the two membranes._____ **Adheres**

42. In the sea urchin, all regions of the egg plasma membrane are capable of fusing with sperm. In several other species, certain regions of the membrane are specialized for sperm recognition and fusion. Fusion is an active process, often mediated by specific "fusogenic proteins? _____**True**

TOPIC NO.53-54

43. The sea urchin egg has ___ mechanisms to avoid polyspermy?_____ **2**

44. The entrance of multiple sperm polyspermy leads to disastrous consequences in most organisms. In the sea urchin, fertilization by two sperm results in a _____ nucleus, in which each chromosome is represented three times rather than twice _____ **Triploid**

45. In normal _____, in which only one sperm enters the egg, a haploid sperm nucleus and a haploid egg nucleus combine to form the diploid nucleus of the fertilized egg (zygote). The centriole, which is provided by the sperm, will divide to form the two poles of the mitotic spindle during cleavage_____ **Monospermy**

TOPIC NO.55-56

46. The Fast block to polyspermy is achieved by changing the electric potential of the egg plasma membrane. This membrane provides a selective barrier between the egg cytoplasm and the outside environment, and the ionic concentration of the egg differs greatly from that of its surroundings?_____ **True**

47. If we insert an electrode into an egg and place a second electrode outside it, we can measure the constant difference in charge across the egg _____ **Plasma Membrane**

48. A third protein released by the cortical granules, a ___, hardens the fertilization envelope by crosslinking tyrosine residues on adjacent proteins?___ **Peroxidase enzyme**

49. The fast sperm inhibition condition is maintained by the plasma membrane, which steadfastly inhibits the entry of…….ions into the oocyte and prevents potassium ions from leaking out into the environment. _____**Sodium**

TOPIC NO.57-58

50. Directly beneath the sea urchin egg plasma membrane is about…cortical granules____**15,000**

51. The removal of sperm is accomplished by the cortical granule reaction, a slower, mechanical block to _____ that becomes active about a minute after the first successful sperm-egg attachment._____ **Polyspermy**

TOPIC NO.59

52. The _____ released by the cortical granules produce an osmotic gradient that causes water to rush into the space between the plasma membrane and the vitelline envelope, causing the envelope to expand and become the fertilization envelope._____ **Mucopolysaccharides**

53. Finally, a fourth cortical granule protein, _____, forms a coating around the egg?_____ **Hyalin**

54. The egg's microvilli attach to…….provides support for the blastomeres ____**hyaline layer.**

TOPIC NO.60-61

55. The release of calcium from intracellular storage can be monitored visually using calcium-activated luminescent dyes such as_____**fura-2**

56. Release of Ca from intracellular storage controlled by-----dye_____ **Aequorin**

57. The drug is …….. calcium ionophore (a compound that transports free calcium ions across lipid membranes)_____ **A23187**

TOPIC NO.62-63

58. Slow block, initiated by the intracellular release of _____ **calcium ions**

59. The fast block, initiated by---------- into the cell in sea urchin_____ **sodium influx**

TOPIC NO.64-65

60. Which has two parts in cell cycle options _____**Blastomere**

61. Transition from fertilization to cleavage is caused by factor____ **MPE**

TOPIC NO.66-67

62. The mitotic division of the nucleus is called_____ **karyokinesis**

TOPIC NO.69

63. Egg activates due to which ion more_____**Ca**

TOPIC NO.70

64. Eggs having sparse and equally yolk space are called _____**isolecithal**

65. Telolecithal is cleavage is found in eggs of _____**Birds and Fishes**

66. The eggs of insects have their yolk in the center_____ **centrolecithal**

67. Divisions of the cytoplasm occur only in the rim of cytoplasm around the periphery of the cell_____ **superficial cleavage**

TOPIC NO.73-74

68. Archenteron is used for _____ **primitive gut**
69. Inner surface cells thin 30 x 5um process _____ **Filopodia**
70. Cells, derived from the micromeres, are called the____ **Primary Mesenchyme.**

TOPIC NO.75-76

71. Spiral cleavage found in ____ **Snails**
72. -----is characteristic of several animals'groups, including annelid worms, some flatworms, and most mollusks (snails)__ **Spiral Holoblastic Cleavage**
73. Blastulae produced by radial cleavage have no blastocoel and are called_____ **Stereoblastulae**
74. Mollusca follow which types of cleavage_____ **Spiral Holoblastic Cleavage**
75. The orientation of the cleavage plane in snail eggs controlled by_____ **Cytoplasmic factor**

TOPIC NO.77-78

76. The snail stereo blastula is relatively_____ **Small**
77. While gastrulating process in snail the micromeres will cover the entire____ **Embryo**
78. Gastrulation in Snails is accomplished primarily by_____ **Epiboly**
79. The resulting larvae of snails is called _ which resemble tiny bear traps __ **Glochidia**

TOPIC NO.79-80

80. Ascidian, members of the phylum _____ **Tunicate**
81. Ascidians are the _____ **Invertebrates**
82. They have a notochord as larvae but they lack _____**Vertebrae**
83. It secretes a ----- tunic (which gave the name "tunicates" to these creatures) ____ **Cellulose**
84. In Tunicate the second cleavage _____ **Meridional cleavage**
85. In Tunicate cleavage during the next three divisions, differences in cell size and shape highlight the _____ of these embryos_____ **Bilateral Symmetry**
86. In Tunicate the second cleavage creates two large interiors blastomeres ___ **A and D**
87. In tunicate the cleavage at 32-cell stage, a small blastocoel is formed and_ begins __**Gastrulation**

TOPIC NO.81-82

88. Tunicate gastrulation is characterized by the invagination of the endoderm. the involution of the mesoderm and the epiboly of the_____ **Ectoderm**

89. Tunicate gastrulation is characterized by the invagination of the _____ **Endoderm**

90. The second step of gastrulation involves the_____ **Involution of The Mesoderm**

91. Pairs of blastomeres appear to lead this invagination into the center of the embryo. This invagination forms a blastopore whose lips will become the --------**Mesodermal cells**

92. The _____ that are the precursors of the neural tube invaginate into the embryo and are enclosed by neural folds _____ **Dorsal Ectodermal Cells**

93. While gastrulation of tunicate the presumptive tail muscle cell ____ are on the posterior lip of the blastopore _____**Yellow crescent**

94. The lateral lips of the blastopore comprise these cells that will become___ **Mesenchyme**

95. The second step of gastrulation in Tunicates involves the involution of the_____ **Mesoderm**

96. The dorsal ectodermal cells that are the precursors of the neural tube invaginate into the embryo and are enclosed by neural folds. This forms the neural tube, which will form a ____ anteriorly and _____ posteriorly_____ **Brain and Spinal chord**

TOPIC NO.83-84

97. Cell fates are determined by signaling. The micromeres constitute a major signaling center. βcatenin is important for the inducing capacity of the ____**Micromeres**

98. --------constitute a major signaling center_____ **Micromeres**

99. A mammalian blastula is called a_____ **Blastocyst.**

100. A_____ is an embryonic structure composed of blastomeres _____ **Blastula**

101. The blastomere cell cycle is governed by the synthesis and degradation of _____ **Cyclin.**

102. A _____is a cell derived from cleavage in an early embryo_____ **Blastomere**

103. In Nematode cleavage is _____ **Rotational**

104. The movements of gastrulation include _____ **invagination, involution, ingression, delamination, and epiboly**

105. Cyclin synthesis promotes the formation of_____ **MPF**

106. Cyclin synthesis promotes the formation of MPF, and MPF promotes_____ **Mitosis**

TOPIC NO.85-86

107. The cytoplasm of the blastodisc becomes the embryo, this type of meroblastic cleavage is called_____ **Discoidal**

108. The first------- divisions occur synchronously, forming a mound of cells that sits at the animal pole of a large yolk cell_____**12**

109. only the cytoplasm of the blastodisc becomes the embryo, this type of meroblastic cleavage is called discoidal present in _____ **fishes**

110. The yolk syncytial layer is formed at the ----------, when the cells at the vegetal edge of the blastoderm fuse with the underlying yolk cell_____ **9th cell cycle**

TOPIC NO.87-88

111. The resulting larvae of snail is called --- which resemble tiny bear traps__ **Glochidia**

112. The first cell movement of fish gastrulation is the --------of the blastoderm cells over the yolk_____ **Epiboly**

TOPIC NO.89

113. The cells adjacent to the chordamesoderm, the paraxial _____ cells, are the precursors of the mesodermal somite_____ **Mesoderm**

114. The cells of both the epiblast and hypoblast intercalate on the future dorsal side of the embryo to form a localized thickening, the _____ **Embryonic shield**

115. After the ---------- cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm_____ **Blastoderm**

116. After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epilogizing blastoderm. This thickening is called the_____ **Germ Ring**

117. The cells adjacent to the chordamesoderm, the paraxial mesoderm cells, are the precursors of the _____ in fish ____ **Mesodermal somite**

118. The neural form _____**Keel Fish**

TOPIC NO.90-91

119. Fertilization of the chick egg occurs in the _____**Oviduct**

120. The first cleavage furrow appears centrally in the_____ **Blastodisc**

121. In birds egg cleavage occurs only in the blastodisc, a small disc of cytoplasm _ in diameter at the animal pole of the egg cell_____ **2-3 mm**

122. The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the _____ **area opaca**

123. Between the area pellucida and the area opaca thin layer of cells called the _____ **marginal belt or marginal zone**

124. Between the blastoderm and the yolk is a space called the_____ **Sub-Germinal Cavity**

TOPIC NO. 92-93

125. The center of this node contains a funnel-shaped depression through which cells can pass into the blastocoels called as_____ **Primitive pit**

126. The major structural characteristic of avian, reptilian, and mammalian gastrulation is the_____ **Primitive Streak.**

127. The depression is called the _____**Primitive Groove**

128. This streak is first visible in birds as a thickening of the _____ at the posterior region of the embryo, just anterior to Koller's sickle_____ **Epiblast**

129. At the anterior end of the primitive streak is a regional thickening of cells called the_____ **Primitive knot or Hensen's node**

TOPIC NO.94-96

130. The mammalian oocyte is released from the ovary and swept by the fimbriae into _____**The Oviduct.**

131. Fertilization occurs in the ------ of the oviduct, a region close to the ovary___ **Ampulla**

132. Fertilization occurs in the ampulla of the_____ **oviduct**

133. The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of_____ **Compaction**

134. In mammalian cleavage at 64-cell stage, the inner cell mass and the _____ cells have become separate cell layers, neither contributing cells to the other group_____**Trophoblast**

135. In mammalian cleavage _____ catches the blastocyst on an extracellular matrix containing collagen, laminin, fibronectin, hyaluronic acid, and heparin sulfate receptors____ **Endometrium**

136. During mammalian cleavage the trophoblast cells secrete fluid into the morula to create a blastocoels this process called as_____ **Cavitation**

TOPIC NO.97-100

137. The mammalian embryo obtains nutrients directly from its _____**Mother**

138. Chorion is derived from _____ cells _____ **trophoblast**

139. This fetal organ the ---- is derived primarily from embryonic trophoblast cells. ___ **Chorine**

140. The fetal portion of placenta formed by _____ **Chorion**

141. Multinucleated type of cell form-_____ **syncytiotrophoblast**

142. Mother provides fetus with _____**Nutrients and Oxygen**

143. It will induce the uterine cells to form the maternal portion of the placenta, the ____ **decidua**

144. While gastrulation process in mammalian _____ becomes rich in the blood vessels that will provide oxygen and nutrients to the embryo_____ **Decidua**

145. The decidua becomes rich in the blood vessels that will provide -------- and nutrients to the embryo. _____ **Oxygen**

146. The cytotrophoblast initially adheres to the ___ through a series of adhesion molecules_____ **Endometri**

147. The narrow connecting stalk of extra-embryonic _____ that links the embryo to the trophoblast eventually forms the vessels of the umbilical cord _____**Mesoderm**

148. The first segregation in mammalian cells within the inner cell mass results in the formation of the ____**hypoblast (sometimes called the Primitive endoderm)**

149. The original type of trophoblast cells constitutes a layer called the__ **Cytotrophoblast**

150. Mother provides the fetus with nutrients and oxygen, and the fetus sends its waste products _____ into the maternal circulation_____ **Carbon dioxide and urea**

TOPIC NO.101-104

151. Fishes,_____, and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg _____**Reptiles**

152. Discoidal meroblastic cleavage occur in_____ **birds and reptiles**

153. The prechordal _____ induces the formation of the forebrain; the chordamesoderm induces the formation of the midbrain, hindbrain, and spinal cord_____ **Plate**

154. Mammals undergo holoblastic ---------cleavage, _____ **Rotational**

155. In birds, ------ is critical in determining the anterior-posterior axis_____ **Gravity**

156. In birds--- differences appear crucial for distinguishing dorsal from ventral __ **pH**

157. Hox genes pattern the anterior-posterior axis and help to specify positions along that___ **Axis**

158. Which forms spinal cord, hind brain and mid brain _____ **Chordamesoderm**

TOPIC NO.105-106

159. Each ICM (Inner Cell Mass) cell has the--------potency_____ **Same**

160. Fraternal twins are the result of ____ separate fertilization events _____**Two**

161. Identical twins share a common _____ **Chorion**

162. Identical twins occur in roughly -------- of human births. _____ **0.25%**

163. About --- of identical twins have two complete and separate chorions___ **33%**

TOPIC NO.107-108

164. The rudiment of the central nervous system, is called _____**Neurulation**

165. Neurulation in _____ is exclusively secondary _____**Fishes**

166. Nervous system develops by _____**Neural Tube**

167. Which type of neurulation in fish _____ **Secondary**

168. In amphibians most of the xenopus most of tadpole is formed by ---------and tail of neural tube formed by 2ndry neurulation _____ **Primary Neurulation**

TOPIC NO.109

169. In primary neurulation, the original ectoderm is divided into ----- sets of cells___ **Three**

TOPIC NO.110

170. The neural folds migrate toward the midline of the embryo, eventually fusing to form the-------beneath the overlying ectoderm_____ **neural tube**

TOPIC NO.113-114

171. In frogs and chicks-------is usually seen in the neural tube of the lumbar (abdominal) and tail vertebrae_____ **secondary neurulation**

172. The growing region at the tip of the lip is called the __**Chord Neural Hinge**

173. Secondary neurulation involves the making of a -------- and its subsequent hollowing into a neural tube. _____ **Medullary Cord**

TOPIC NO.115-118

174. Hindbrain_____ **Rhombencephalon**

175. One agent of ventral specification is the ---------hedgehog protein_____ **Sonic**

176. Sonic hedgehog is a type of_____ **Protein**

177. The telencephalon will eventually form the cerebral hemispheres, and the diencephalon will form the thalamic and -------brain regions_____ **Hypothalamic**

178. Retinoic Acid comes from _____ **Somite's**

TOPIC NO.119-120

179. Mantle cells differentiate into_____ **Gray Matter**

180. The mantle zone cells differentiate into both _____**Neurons and Glia**

TOPIC NO.121-122

181. Marginal zone to form clusters of neurons called_____ **Nuclei**

TOPIC NO.123

182. The human brain consists of_____ **10^{11} neurons**

183. Human brain contains _____**10^{12} Glial Cells**

184. Those cells that remain integral components of the neural tube lining become----- cells _____ **Ependymal**

TOPIC NO.124

185. Dendrites are often numerous and do not extend far from the neuronal cell body, or----------, axons may extend for several feet. _____ **Soma**

TOPIC NO.125-126

186. The cells at dorsal most portion of neural tube become _____ **Neural Crest Cells.**

187. The neural crest cells originate at the------- most region of the neural tube. _____ **Dorsal**

188. Neural crest divides functionally indomains_____**4**

TOPIC NO.127-130

189. ---------- plays a major role in eye formation_____**Pax6**

190. Paracrine factors such as --- and FGF7 are important in normal skin development. ____ **TGF**

191. Paracrine factors such as TGF- and FGF7 are important in normal ___**Skin Development.**

192. The cerebral cortex in humans has----------layers. _____ **Six**

193. TGF-β family secreted from the surface ----- and roof of the neural tube. ____ **Ectoderm**

194. The space between two neurons which transmit are released_____ **Synapse**

TOPIC NO.131-132

195. Paracrine factors instruct the ---------- to become muscles by inducing them to synthesize the Myoid protein_____ **Myotome Cells**

196. Paracrine factor instruct myotome cell to become_____ **Muscles**

197. Myogenic bHLH proteins are specific for development of _____**Muscle Cell**

TOPIC NO.133-134

198. The second step is the ------of the myoblasts together into chains. __ **Alignment**

199. When myoblasts stop dividing, secrete _____**Fibronectin**

TOPIC NO.135-136

200. There are-------major modes of bone formation, or osteogenesis _____**Two**

201. The direct conversion of mesenchymal tissue into bone is called _____**Intramembranous Ossification**

TOPIC NO.137-138

202. The osteoblasts secrete a collagen-proteoglycan -- that is able to bind calcium salts. __ **Matrix**

203. The entire region of calcified spicules becomes surrounded by ------ mesenchymal cells that form the Periosteum _____ **Compact**

TOPIC NO.139-142

204. Endochondral ossification involves the formation of _____**Cartilage**

205. Mutations of the -------- cause campomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body_____**SOX9 Gene**

206. Mutations of the------cause camptomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body____ **SOX9 gene**

207. ------- and Scleraxis are the transcription factors are thought to activate cartilage-specific genes ____ **pax1**

208. Scleraxis is expressed in the------from the sclerotome_____ **mesenchyme**

209. ------phase involves the invasion of the cartilage model by blood vessels _____**Fifth**

210. The SOX9 gene, which ------ a DNA-binding protein, _____ **Encodes**

TOPIC NO.143-144

211. Osteoclasts are probably derived from the ---- precursors as macrophage blood cells__ **Same**